

Chronic Kidney Disease of Uncertain Etiology

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KEY POINTS

- A number of geographic regions around the world have an unexpectedly high prevalence of kidney disease among persons without previously identified risk factors for kidney disease.
- Agricultural work is the predominant occupation in three of the best-studied hotspots: Pacific-Coast Central America, North Central and Central Province, Sri Lanka, and Uddanam region within Andhra Pradesh, India.
- Kidney biopsy series, though small in number, indicate tubulointerstitial disease with variable degrees of glomerulosclerosis across all three regions.
- Accurate assessments of prevalence and incidence are challenging, in part due to resource limitations but also due to lack of clear demographic or clinical (laboratory-based) parameters distinguishing chronic kidney disease from chronic kidney disease of unknown etiology.
- Several hypotheses regarding cause are under study, including recurrent heat stress; exposure to ground water toxin; or infection. The investigation of these hypothesized causes necessitates collaborations across occupational health, environmental epidemiology, and nephrology, in addition to a number of large international studies building scientific resources and capacity in otherwise underresourced regions.

Since the latter part of the 20th and early 21st century, nephrologists have identified a progressive kidney disease, primarily tubulointerstitial on kidney biopsy, occurring in specific regions of South Asia and Central America, and among persons without previously identified risk factors for kidney disease. Much remains unknown and unclear about this disease, starting from nomenclature. In this chapter we use the term “Chronic Kidney Disease of Uncertain Etiology (CKDu)” but the terms Meso-American endemic Nephropathy (MeN), Uddanam Nephropathy, and Chronic interstitial nephritis in agricultural communities (CINAC) are also commonly used. “CKDu” is an inherently broad and opaque term. In this chapter, we use CKDu to refer to the highly-prevalent kidney disease in specific agricultural

regions, disproportionate to the age distribution and diabetes prevalence, and in which kidney biopsies, where conducted, have demonstrated primarily tubulointerstitial kidney disease. We distinguish this entity from many other patients with “unknown” CKD, which could occur due to late presentation or lack of access to diagnostic testing, and other known causes of tubulointerstitial kidney disease such as hypersensitivity reaction to antibiotics or autoimmune disease such as sarcoidosis.

A number of other “endemic” regional nephropathies might historically have met criteria for CKDu until the cause was identified. These include unregulated release of cadmium into the Jintsu river during World War II in Japan, leading to contaminated rice paddy fields and an outbreak

of “itai-itai” disease;¹ lead nephropathy due to lead-based paint used in verandas in Queensland, Australia;² and aristolochic acid nephropathy in the Balkans in the 1940s where a nephrotoxic weed growing in the wheat fields was admixing with families’ flour.³ Tubular injury predominated as the primary finding on biopsy in all three of these regional nephropathies. In the latter case of Balkan endemic nephropathy, although the timeline to identification of the cause was prolonged—nearly 6 decades—eventually definitive evidence emerged to identify chronic dietary exposure to aristolochic acid as the cause.³ Its discovery was serendipitously facilitated by a 1990s “outbreak” of rapidly progressive renal failure with tubulointerstitial nephritis among women attending a weight-loss clinic in Belgium.⁴ These women had ingested Chinese herbs, iterative analyses of which yielded the presence of aristolochic acid. Subsequent translational work identified aristolochic acid DNA adducts in affected patients’ kidney tissue or urine.⁵ Research in CKDu cause(s) is yet to identify such convincing evidence linking epidemiology to pathology and thereafter to pathophysiology, as was eventually described for Balkan endemic nephropathy.

FIRST DESCRIPTIONS IN THREE HOTSPOT REGIONS

FROM BAJO LEMPA IN EL SALVADOR TO OTHER COASTAL REGIONS IN CENTRAL AMERICA

Lacking any systematically collected end-stage kidney failure or CKD registry data, a rise in kidney disease–related hospitalizations alerted nephrologists in El Salvador to an unusual kidney disease occurring in their regions. A 1999 study in El Salvador noted that two-thirds of these patients exhibited a peculiar profile, distinguishing them from that of the classic patients with CKD: they were relatively young (mean age 51 years), mostly male (87%), agricultural workers (63%) from specific coastal regions and lacked history of diabetes or hypertension (96%).⁶ Several subsequent studies confirmed a similar profile^{7,8} (Fig. 83.1). Most patients seemed to come from the basin and the mouth of the Lempa River; hence the epidemic was referred to as the “Bajo Lempa nephropathy” in El Salvador (Fig. 83.2). Simultaneous observations and subsequent studies demonstrated the presence of the disease in other geographic hotspots along the whole Pacific Ocean Mesoamerican coastline from southern Mexico to Panama, also comprising Guatemala, El Salvador, Nicaragua, and Costa Rica.^{9,10} Many researchers and nephrologists began to group these patients together using the term “Mesoamerican endemic nephropathy (MeN) (see Chapter 74).¹¹”

Later studies described the clinical profile of the disease at earlier stages. Patients were described as presenting with cramps, paresthesia, muscle weakness, and occasionally with aseptic dysuria.^{7,9,11–14} Presentation with an acute kidney injury (AKI) event has also been described, with accompanying fever and pyuria, with limited long-term follow-up data.^{15,16} Blood pressure is normal referent to population averages, proteinuria is minimal or absent (except in advanced stages, when secondary glomerulosclerosis may be substantial), and hyperuricemia

and electrolyte disturbances, especially hypokalemia and hyponatremia, are common. A family history of CKD is also common. In advanced stages, patients often present with elevated serum creatinine levels, variable proteinuria, important anemia, and uremia, but still with normal or only slightly elevated blood pressure, normal or low potassium levels, and absence of edema or even dehydration. Yet no single presenting feature is considered pathognomonic among clinicians practicing in the region, and the diagnosis is generally established by ruling out other common causes of CKD and/or pursuing kidney biopsy.

NORTH CENTRAL AND CENTRAL PROVINCE, SRI LANKA

Mirroring the story from El Salvador, in the mid-1990s, alarmingly high numbers of young farmers from rural communities in the North Central Province of Sri Lanka presented to the regional tertiary care hospital with uremic emergencies. There was no identifiable cause in the majority of these patients, who commonly presented with kidney failure requiring acute peritoneal dialysis as the only available treatment option.¹⁷ Comparing data from two regional hospitals, the number of hospitalizations for kidney failure due to unknown reasons was found to be highest among people residing in the North Central Province of the country.¹⁸ A population-based study of >6000 persons in three different geographical locations of the country reported on the prevalence of dipstick-detected proteinuria. Although the prevalence of dipstick-positivity was <10% in all three regions, in the surveyed North Central region a large proportion (87%) of those with dipstick-detected proteinuria did not have an identifiable cause, compared with a cause being identified in >90% with dipstick-detected proteinuria in the two other surveyed regions.¹⁷

Early efforts relied on dipsticks,¹⁷ but given that the pathologic lesion of the disease is primarily tubulointerstitial, such screening efforts may have lacked sensitivity and specificity.¹⁹ Thereafter several waves of community screening and detection efforts have occurred in Sri Lanka, with variable success. Since a majority of the rural population relies on government-funded health care, which is integrated across primary to tertiary care, one report used hospital- and clinic-based case records to map the geographic distribution of CKD (Fig. 83.3). Population-based data surveying the expected low and high prevalent regions based on this map indicate a prevalence of estimated glomerular filtration rate (eGFR) of <60 mL/min/1.73 m² of up to 12% among men without diabetes or hypertension in the North Central Province.²⁰ A biopsy study demonstrated that primary tubulointerstitial kidney disease was present in nearly half of kidney biopsies performed in the major tertiary center in the Central Province.²¹

Some clear similarities exist in the demographic and clinical characteristics of Sri Lankan and Central American patients: a male predominance among those affected, reports of significant hypokalemia in the early stages, and a subset may present acutely with systemic illness.^{15,16} Notable differences include a biochemical profile with hypernatremia and lack of significant hyperuricemia in Sri Lanka compared with Central America.^{22,23} Table 83.1 summarizes some similarities and differences in clinical

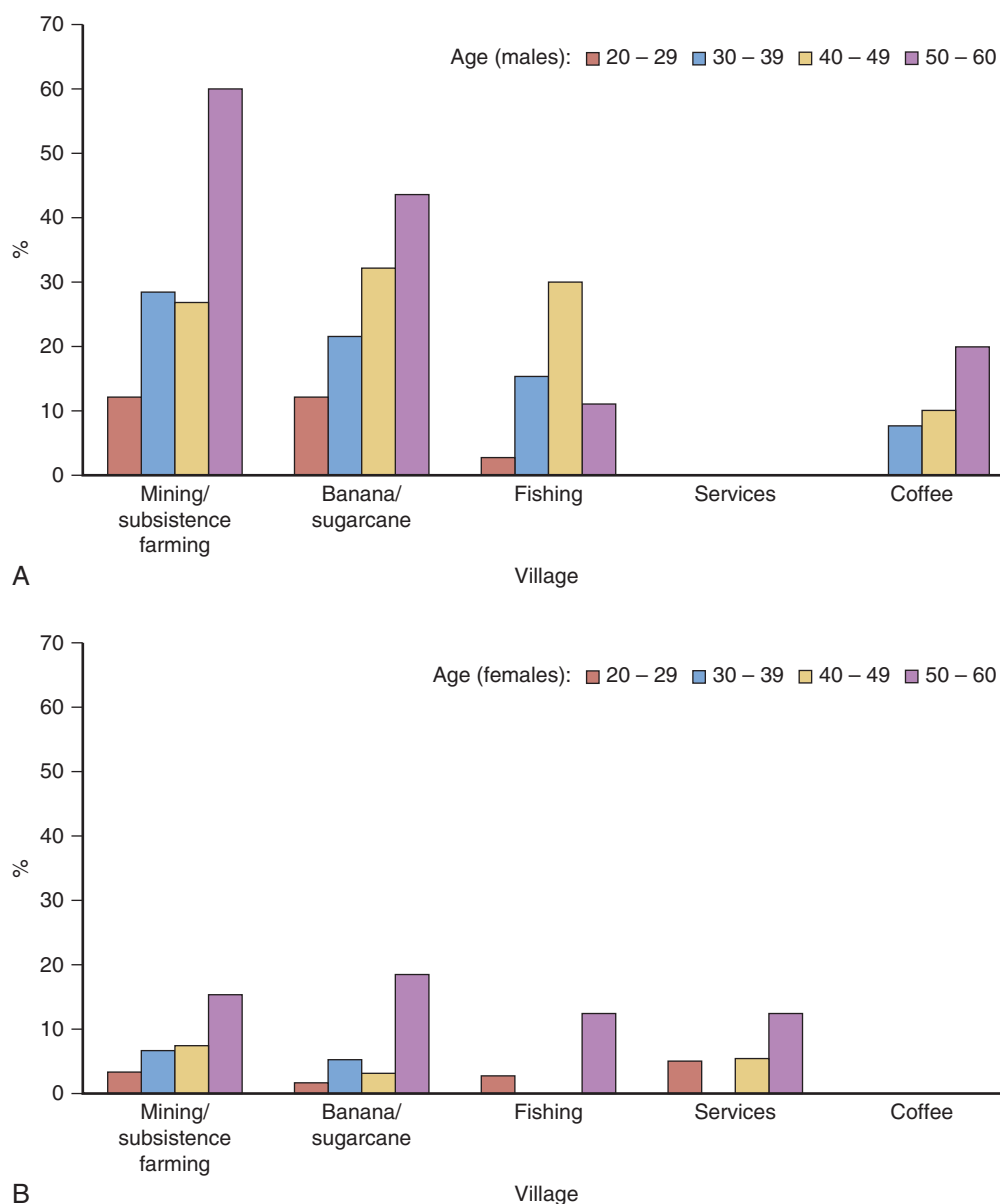


Fig. 83.1 (A [males] and B [females]) Examples of early epidemiologic studies evaluating chronic kidney disease of unknown etiology in Central America. Prevalence of abnormal serum creatinine (>1.2 mg/dL for men, >0.9 mg/dL for women) in 1096 residents aged 20 to 60 years in 5 villages in northwestern Nicaragua. Villages are grouped by primary occupation. (From Torres C, Aragon A, Gonzalez M, et al. Decreased kidney function of unknown cause in Nicaragua: a community-based survey. *Am J Kidney Dis*. 2010;55(3):485–496.)

presentation and associated features on the basis of existing data from the two best-described hotspots.

There is an interesting historical context to CKDu-endemic regions in Sri Lanka. Before 1970s, the CKDu-endemic areas in Sri Lanka were sparsely populated. Several major irrigational projects were launched to divert water from the wet zonal hill country to the north central and central dry zones for enhancing agriculture to ensure food security of the country. In parallel, there was intentional large-scale population migration from wet zonal areas to these dry zonal areas of the country. The CKDu epidemic emerged approximately 2 decades later. This historical timeline has heightened suspicions of environmental exposure, specifically water-based exposure, as a potential contributor to the epidemic.

UDDANAM REGION, INDIA

Another major recognized hotspot of CKDu is in southern India, termed “Uddanam endemic nephropathy” for the specific region within Andhra Pradesh, where an endemic kidney disease was first recognized.^{24–26} Uddanam is a coastal region of this state, also relying primarily on agriculture but with a greater variety of crops including coconut, cashews, rice paddy, and vegetable cultivation (Fig. 83.4) (see Chapter 77).

A large cross-sectional study of more than 2400 persons from this region indicated nearly 20% of men and 16% of women had CKD, a majority ($>70\%$) were engaged in farm work, diabetes was present in less than 20%, and family history of kidney disease was reported in 25% of the subset with $\text{eGFR} < 60$ mL/min/ 1.73 m².²⁷ A prior study had

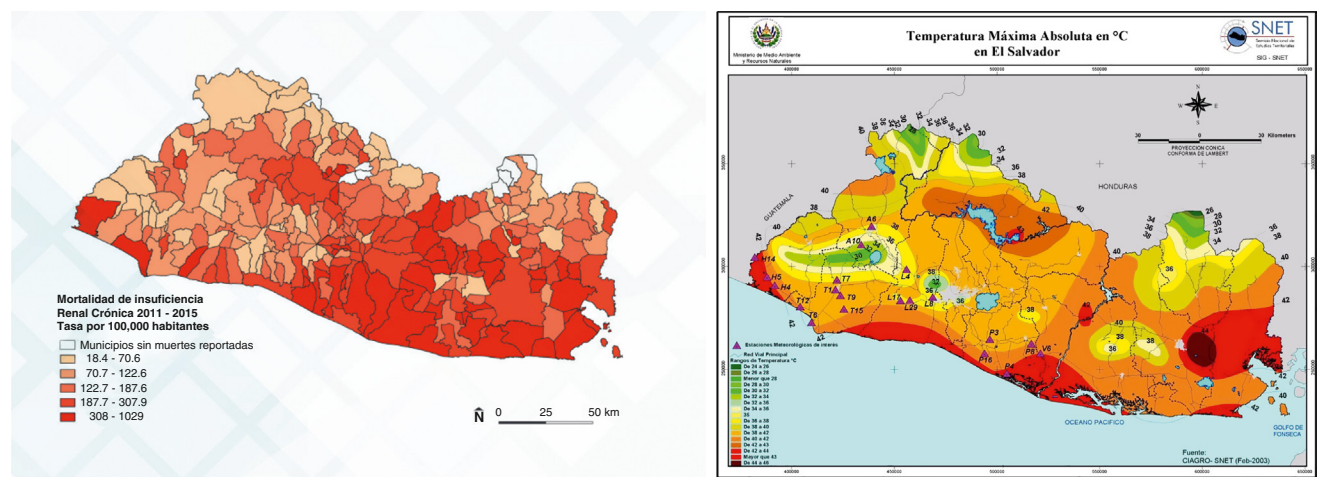


Table 83.1 Similarities and Differences in Clinical Presentation and Other Associated Features of Chronic Kidney Disease of Unknown Etiology in the 2 Best-Described Hotspots

	Mesoamerica	Sri Lanka
Geospatial Factors		
Hot, dry climate	✓	✓
Reliance on groundwater supply	✓	✓
High-density agrochemical application	✓	✓
Low population-based prevalence of diabetes	✓	✗
Clinical Presentation/Other Associated Features		
Male predominance	✓	✓
Lack of heavy proteinuria	✓	✓
Significant hyperuricemia	✓	✗
Significant hypokalemia	✓	?
Extensive use of NSAIDs	✓	✗
Specific job tasks with higher risk	✓	✗
Common tobacco use	?	✓
Biopsy Data		
Chronic tubulointerstitial nephritis and fibrosis	✓	✓
Some with primarily acute interstitial nephritis	✓	✓
No immune complex deposition	✓	✓
Significant glomerulosclerosis	✓	?

Question mark symbols indicate lack of definitive data to make a determination.
NSAIDs, Nonsteroidal antiinflammatory drugs.

reported similar findings.²⁸ Notably, in the earlier study of 2200 persons, nearly 50% of persons with CKD had urine-to-protein creatinine ratios <0.2 g/g. A small case series confirmed that the primary findings among five men and one woman on biopsy were predominant tubulointerstitial lesions with a variable degree of glomerulosclerosis, with serum creatinine levels <2 mg/dL in the majority.²⁹

In a recent follow-up study using verbal autopsies in the region, CKD was identified as the primary cause of death among 133 deaths, accounting for 45% of all deaths in the region.³⁰ For context, in the Million Death Study, conducted across all of India, CKD was determined to be the cause of 2.9% of the deaths among persons 15 to 69 years of age.³¹

CHALLENGES TO DETECTION EFFORTS

Despite substantial coverage in the popular media, robust data on the burden of CKDu globally remain sparse outside of a few intensely studied regions in Central America and South Asia. In part, this reflects the challenges of capturing estimates of disease in the populations affected, which are

typically served by poorly developed health systems and which have inadequate access to care (including kidney replacement therapy). This means routine health record data sources, such as death certificates or dialysis registries, can only provide limited insights into the disease burden and risk factors in most regions.³²

Efforts to characterize the prevalence of CKDu using population-representative surveys are ongoing.^{33,34} These studies are resource intensive as eGFR testing based on blood sampling is required to define cases, given that urinary abnormalities are not typical in CKDu. eGFR test values can in turn be challenging to interpret as alongside interlaboratory variability,³⁵ neither eGFR estimating equations nor the healthy population distribution of the measured GFR are generalizable across regions. The epidemiologic study designs used typically include only a single measure of eGFR without confirmation of chronicity, which may overestimate CKD prevalence.³⁶ These studies also have limitations surrounding (often variable) case definitions for CKDu. Differentiating cases of CKDu from other causes of CKD is challenging even at the individual level, and pragmatic definitions such as the absence of significant proteinuria, hypertension, and diabetes (and sometimes retroviral disease and nephrotoxic drug use) are necessary in epidemiologic studies. These factors all inevitably lead to misclassification because 1. not all CKD is CKDu even in endemic areas; 2. both diabetes and hypertension, although not features of the disease, can clearly coexist with CKDu; and 3. there are cases of nonproteinuric CKD (e.g., due to drugs and genetic disorders or of unclear etiology) in all regions (i.e., including high-income nonagricultural settings), although this constellation likely does not represent the same clinical entity as the “CKDu” disease(s) described in this chapter. Nonetheless, assuming the contribution of any misclassification is low, as will be the case where there are high levels of CKDu, these studies can still provide valuable insight into the burden of, and risk factors for, CKDu across different regions globally. A number of population-representative studies conducted in areas of reported high CKDu burden, or with the specific aim of studying CKDu prevalence, are summarized in Table 83.2.

Due to the differences in population age structures and the other limitations described earlier, the results of the above studies are not directly comparable, although a number of common themes can be identified.

Firstly, where CKDu epidemics are reported, population rates of eGFR <60 mL/min are typically ≥5% when those with diabetes, hypertension, and proteinuria are excluded and chronicity is not confirmed. Estimates would be expected to be 1% to 2% or lower using the same criteria (assuming similarly young populations) in non-CKDu-endemic regions.

Second, a number of risk factors are common across these studies. These include age (which is a risk factor for all forms of CKD), male sex, lower altitude, tobacco smoking, and lower levels of education/higher levels of poverty. Finally, agricultural (specifically sugarcane) work, rural residence, or residence in an agricultural community are also typically identified as risk factors, although distinguishing the occupational from residence-related risk factors is not always possible in many of these studies.

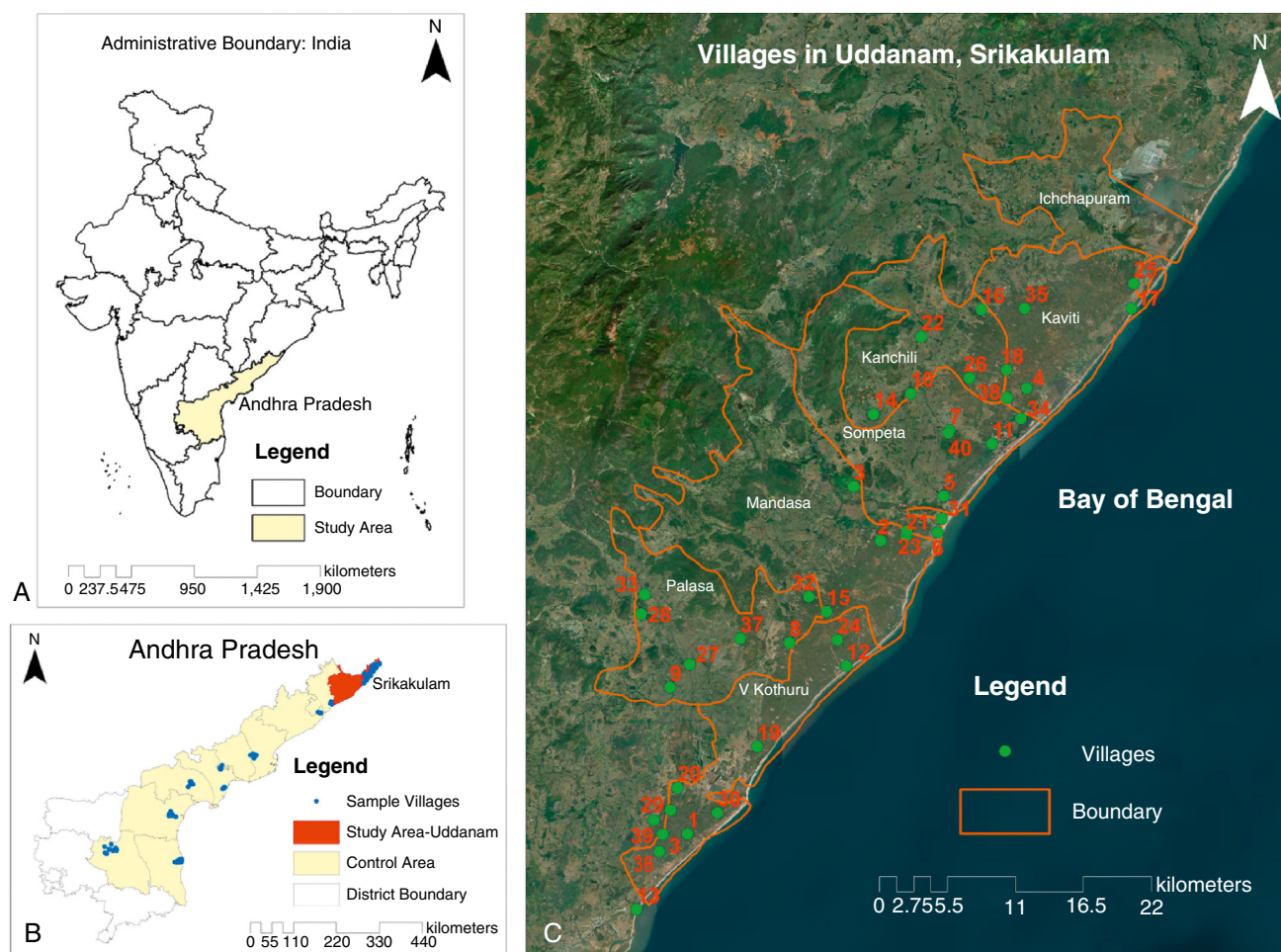


Fig. 83.4 Investigations in chronic kidney disease of unknown etiology in the Uddanam region, Andhra Pradesh, India. Uddanam region in the Southern state of Andhra Pradesh is an identified “hotspot” of chronic kidney disease of unknown etiology. Green dots indicate sampled villages. This study compared groundwater samples from the high-prevalence region to geographically similar areas and noted a higher prevalence of acidic, hard water with higher silica content but no evidence of significant pesticide contamination. (From Lal K, Sehgal M, Gupta V. Assessment of groundwater quality of CKDu affected uddanam region in Srikakulam district and across Andhra Pradesh, India. *Groundwater Sustain Develop.* 2020;11:100432.)

BIOPSY FINDINGS

As with previous endemic nephropathies, kidney biopsy studies in these current CKDu-endemic regions are limited but consistent with a tubulointerstitial disease, showing mainly tubular atrophy with interstitial fibrosis and glomerulosclerosis related to ischemia, without immune complexes^{37,38} (Fig. 83.5). One of the first case series describing pathology findings reported data from eight men with serum creatinine ranging from 1.1 to 2.7 mg/dL, and this case series described both significant interstitial fibrosis and glomerulosclerosis. In fact, tubulitis and interstitial inflammation, although described, were not severe.³⁹ However, studies from both Sri Lanka and Nicaragua recruited patients with “acute symptoms” (e.g., dysuria, fever, and AKD) from CKDu-endemic regions and showed features of acute interstitial nephritis (i.e., tubulitis and inflammatory infiltrate in the tubules^{12,15,16,21}), with changes in the glomeruli confined to areas with chronic interstitial damage. Assuming this clinical presentation represents the onset of CKDu, these studies would favor the argument that a primary tubulointerstitial disease leads

to secondary glomerulosclerosis but certainly do not rule out the possibility that an ischemic insult is driving both tubulointerstitial and glomerular injury in parallel.

Deeper review of tissue samples has brought a finding of potential interest: electron-dense lysosomal granules in the tubules.^{40,41} The authors hypothesized that these granules may indicate an injury pathway similar to that instigated by calcineurin inhibition. Of note, however, is that replication studies have demonstrated these findings to be relatively nonspecific as others were able to identify similar granules in persons with known glomerular disease.⁴² Additional electron microscopic findings described include patchy podocyte foot process effacement, tubular vacuolization, and mitochondrial swelling.

Several challenges exist in describing the spectrum of kidney pathology observed with CKDu. A small sample of persons with CKDu undergoes biopsy, with quantitatively fewer biopsies available from Central America than from Sri Lanka. In Central America, patient-borne costs are higher. Lack of dedicated pathologists may present a further impediment. Even in persons with biopsies, there is tremendous heterogeneity in the level of kidney dysfunction

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies

Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demon-strated	% eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
O'Donnell et al. ⁸⁵	2011	Central America	Leon, Nicaragua	771	N but low prevalence	N	N		N	13%	Age Male sex Low altitude
Torres et al. ¹⁰	2010	Central America	Leon and Chinandega, Nicaragua	1096	N but low prevalence	N	N		N	8.7%	Age Male sex Agricultural or mining communities
Peraza et al. ⁸⁶ https://doi.org/10.1053/j.ajkd.2011.11.039	2012	Central America	Pacific Coast, El-Salvador	664	N but low prevalence	N	N		N	0% in high altitude/urban areas 10% in sugarcane communities	Age Male sex Low Altitude Sugarcane/cotton work Tobacco use (males only) Kidney stones (males only) Diabetes (females only)
Orantes et al. ⁸⁷	2011	Central America	Bajo Lempa, El-Salvador	775	Y	Y	N		Y	3.6%	Hypertension (females only) Age Male sex Family history of CKD
Miller et al. ⁸⁸ https://doi.org/10.1016/j.ekir.2020.12.015	2021	Central America	Tecpán, Guatemala and San Antonio Suchitepéquez, Guatemala	807	N	N	N		N (Sensitivity analysis only)	4.0%	Age Diabetes Hypertension Low BMI Sugarcane work in poorer participants

Continued on following page

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies (Continued)

Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demonstrated	% eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
Ruiz-Alejos et al. ⁸⁹	2021	South America	Tumbes, Peru	1,514	Y	Y (140/90)	Y		N	0.9%	Numbers too small to undertake adjusted analyses in those without hypertension or diabetes
O'Callaghan-Gordo et al. ⁹⁰	2019	South Asia	Chennai, Visakhapatnam, Haryana and Delhi, India	12,500 (after exclusion of diabetes and hypertension)	Y	Y (140/90)	Y		N	1.4% Urban Northern areas Rural residence 1.9% Rural (Southern areas only) 0.43% Urban Fewer years of Southern area education 4.8% Rural Southern area 13.3%	Age
Tatapudi et al. ²⁸	2019	South Asia	Uddanam, India	2210	Y	Y (140/90 for more than 5 years)	Y (PCR >1000 mg/g)		N		Age
Gummididi et al. ²⁷	2020	South Asia	Uddanam, India	2419	N	N	N		Y	10.2%	Age Male sex Hypertension Tobacco use Lack of formal education Outdoor work Family history of CKD
Ruwanpathirana et al. ²⁰	2019	South Asia	Anuradhapura district, Sri Lanka	4803	Y	Y (140/90)	Y (ACR >300 mg/g)			3.7% in lowest impacted area 8.5% in highest impacted area	Analgesic use Age Male sex Family history of CKD Tobacco use (males only)

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies (Continued)

Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demonstrated	% eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
Jayatilake et al. ⁹¹	2013	South Asia	Anuradhapura, Badulla, Hambantota and Polonnaruwa districts, Sri Lanka	4957	Y	Y (160/90)	N		N	NA	Proteinuria screening only
Hathaway et al. ⁹²	2023	Sub-Saharan Africa	Kisumu, Kenya	782	Y	Y	Y	HIV	N	0%	NA
Hamilton et al. ⁹³	2020	Sub-Saharan Africa	Lilongwe and Karonga district Malawi	821	Y	Y (140/90)	Y			0.2%	NA
Aekplakorn et al. ⁹⁴	2023	Southeast Asia	National, Thailand	17,329	Y	Y (140/90)	Y (urinalysis >1+)	Age >70	N	0.7% Central, South and Bangkok regions 1.6% Northeast region	Age Female sex Gout Kidney stones Rural residence Analgesic use Manual labour/agricultural work Low BMI

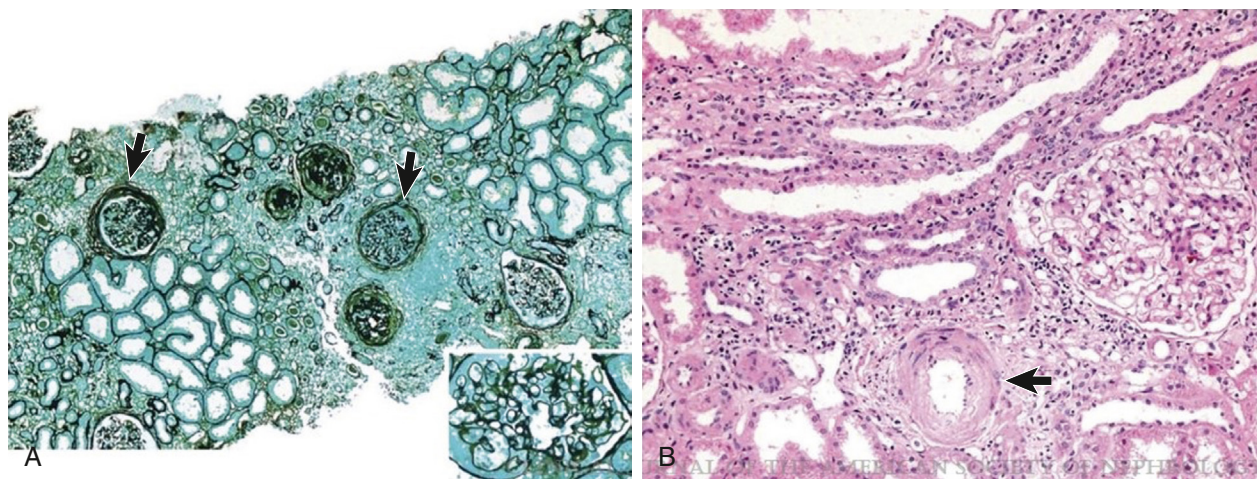


Fig. 83.5 Kidney biopsy findings from a tertiary care hospital in Sri Lanka. (A) Broad bands of tubular atrophy and interstitial fibrosis with relative sparing of glomeruli. In addition to global glomerulosclerosis, this atrophic cortex shows glomeruli with periglomerular fibrosis (arrows). The preserved glomeruli are unremarkable (inset), and the immunofluorescence microscopy is negative (methanamine silver stain, $\times 100$). (B) Active interstitial inflammation in nonatrophic cortex. The infiltrate is composed of lymphocytes admixed with a few plasma cells and rare eosinophils. The glomerulus is normal, and the interlobular artery (arrow) shows mild arteriosclerosis (hematoxylin-eosin, $\times 200$).

at the time of biopsy. Thus in the existing literature, we lack sufficient numbers of cases to confidently match level of kidney dysfunction to tissue findings.

Developing molecular interrogation techniques holds promise for identification of pathognomic injury pathways (which could in turn provide clues to potential etiologies). However, application of these techniques (e.g., single-cell RNA sequencing) requires resource-intensive tissue handling and storage, further complicated by the need to transfer tissue from regional hotspots to the specialized centers with technical expertise.

HYPOTHESIZED CAUSE(S) OF CKDU

Despite intensive efforts for more than 2 decades, some of which have garnered well-deserved attention by the lay press, as of this chapter's writing no specific etiology is widely accepted as the primary driver of disease. The hypothesized causes span multiple domains including genetic, infectious, occupational, and environmental (Fig. 83.6). The prolonged struggle to eliminate or deprioritize any of these broad domains reflects, among other factors, the failure to identify early kidney disease, matched with lack of methodologies to elicit cumulative (lifetime) measures of exposures (Fig. 83.7). This in turn complicates prospective investigations which need to continue to expend resources to interrogate multiple domains.

Several major hypotheses have emerged as to the cause of CKDu, and we have created a comprehensive, though not exhaustive, list with supportive data (Table 83.3). Among this list, occupational heat stress has been a major focus in Central America. Specific areas affected within India, Sri Lanka, and Central America are some of the hottest areas within these regions (see Fig. 83.2). Furthermore, populations affected by CKDu often undertake manual labor in hot conditions. In Central America these include sugarcane farming, subsistence farming, brick making, or construction;^{43,44} in Sri Lanka and India these include rice

farming, sand mining, or cashew farming.⁴⁵ In Sri Lanka and India, water contamination by metals or metalloids, exposure to agrichemicals, and infection have received more attention as potential causes of disease.

HEAT STRESS AS A CAUSE OF CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

In Central America established CKD prevalence is high among sugarcane workers and other manual occupations.^{16,17,44,46} Given the associations with hot work and reduced kidney function in workplace studies (see later), it is hypothesized that heat exposure and physical exertion without adequate hydration leads to repeated tubular injury and subsequent fibrosis.⁴⁷ Specifically, it is hypothesized that a rise in core metabolic temperature, which relates to not only the ambient temperature but also the intensity of the metabolic task being performed, results in AKD, repeated episodes of which lead to eventual chronic kidney disease. Putatively, since the kidney medulla in particular is one of the most metabolically active environments in the body while also operating within a narrow arteriovenous oxygen gradient, it could be sensitive to either vasodilatory hypotension or volume depletion. Additionally, proinflammatory responses that among other effects activate the clotting cascade could further either directly injure renal parenchyma or lead to a systemic inflammatory response syndrome.⁴⁸

Some of the most detailed physiologic studies describing the kidney-specific response to heat stress were performed among healthy U.S. military recruits undergoing summer training in the 1970s.⁴⁸ In these studies the combination of heat exposure and strenuous exercise was associated with dramatic changes in the urine (90% of recruits had pyuria after ~40 days of training) and modest changes in serum creatinine (mean change from 1.0–1.3 mg/dL) and uric acid; the latter two markers did not change among recruits undertaking moderate exercise.^{48,49} Creatinine clearance did not change among recruits undertaking moderate

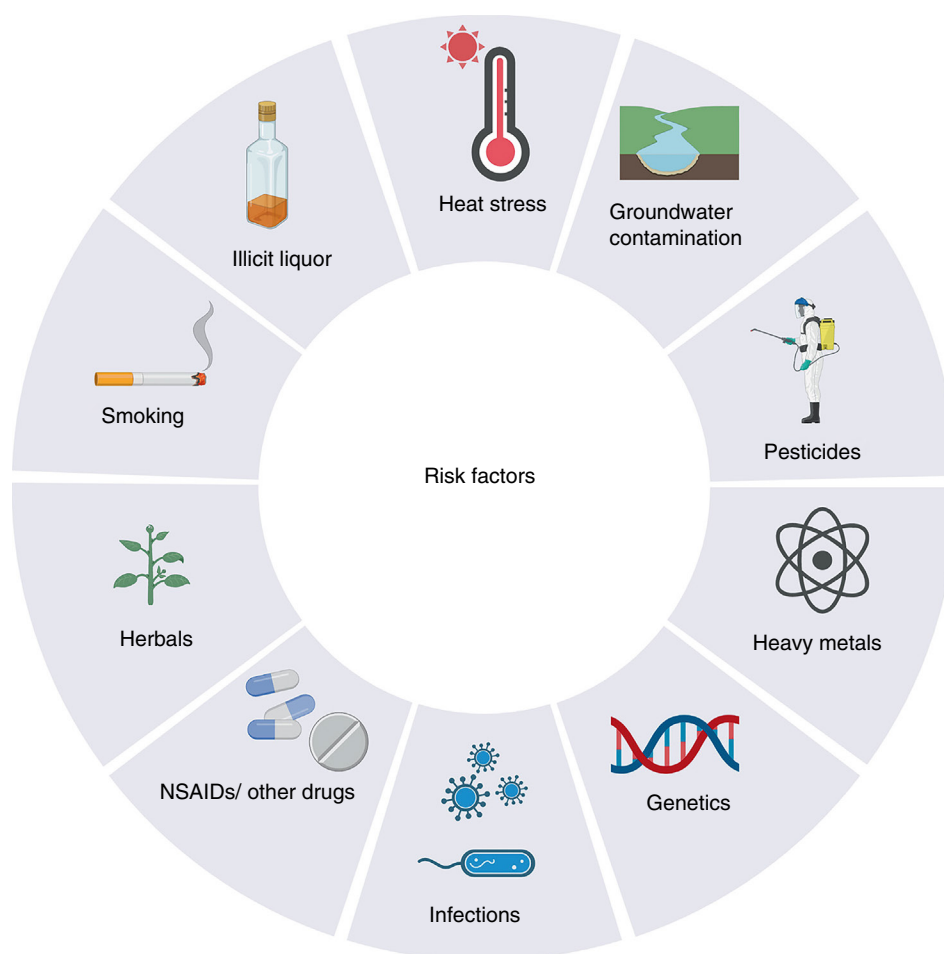


Fig. 83.6 Hypothesized risk factors for chronic kidney disease of unknown etiology (CKDu). The hypothesized causes of CKDu are many and include multiple domains, the measurement of which is time consuming and resource intensive. (From Rao IR, Bangera A, Nagaraju SP, et al. Chronic kidney disease of unknown aetiology: A comprehensive review of a global public health problem. *Trop Med Int Health*. 2023;28(8):588-600.)

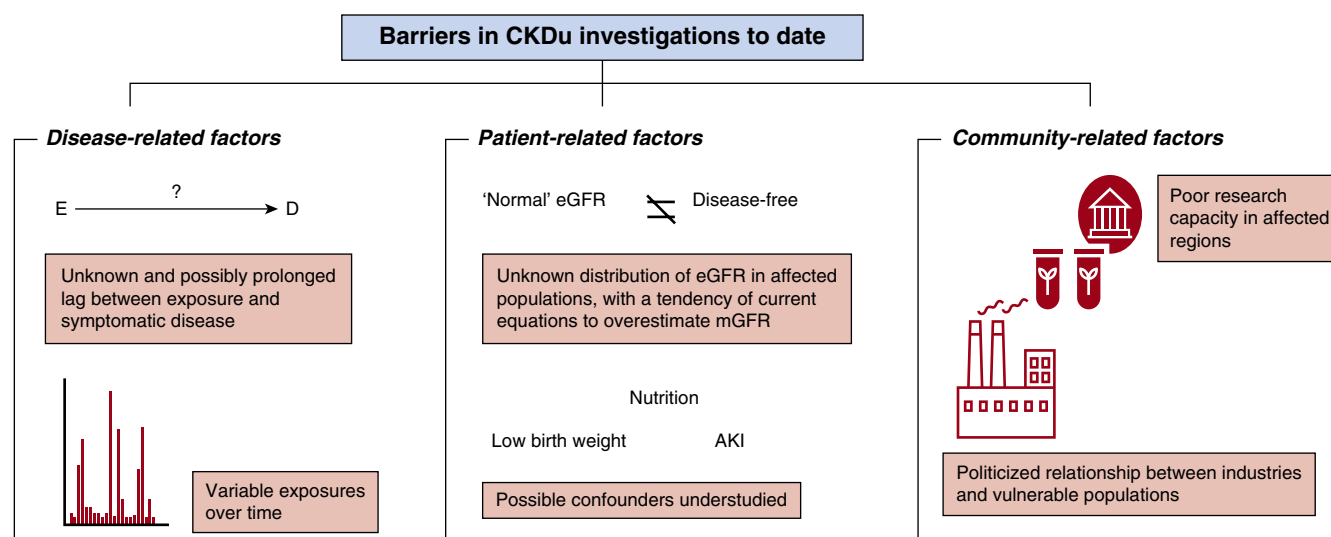


Fig. 83.7 A schema of barriers to investigating cause(s) of CKDu. (Adapted from a figure generated by Winne Ellerman.)

exercise, but it was not performed for those with more strenuous training. Finally, plasma osmolality increased and urine osmolality declined, although urine to plasma sodium remained stable.

The current body of evidence surrounding heat stress and kidney injury, whether epidemiologic or pathophysiologic, mainly interrogates acute or subacute kidney injury in context of heat stress experienced during strenuous activity.

Table 83.3 Hypothesized Potential Cause(s) of Chronic Kidney Disease of Unknown Etiology (CKDu)

Proposed Causes of CKDu	Example Supportive Evidence
Occupational heat exposure	Cross-shift change in serum creatinine ^{68,95} Cross-harvest change in eGFR among those with most strenuous tasks ^{69,72}
Metals and metalloids	Fluoride ⁵¹ High overall mineral content ("hard water") ^{45,51} Nickel ⁵³ Silica nanoparticles ⁵⁴ Other metals including arsenic
Agrochemicals	Glyphosate ^{74,96} Paraquat ⁹⁷ Mixture of agrochemicals through well water ^{45,96}
Infections	Leptospirosis ⁵⁷ Hantavirus ⁵⁶ Cyanobacteria ⁹⁸
Genetic	Proximal tubular potassium channel polymorphism ⁶⁰

Longer-term prospective studies on job-task differentiated risks for chronic kidney disease are ongoing.⁴⁴ Such studies are rare because they encounter tremendous logistical barriers when workers disperse to far-flung rural regions post harvest. Furthermore, in Sri Lanka and India, job tasks are not well defined as many farmers own or lease entire plots of land and complete all tasks associated with the harvest season.²¹

Arguments against occupational heat exposure as a *primary* cause of disease include the lack of described high prevalence of kidney dysfunction among similar occupational settings in countries outside Central America, India, and Sri Lanka, alongside the strong association described between ambient heat exposure and kidney disease progression in other forms of CKD (e.g., CKD due to diabetes or glomerulonephritis).⁵⁴ It has therefore been suggested that the well-described associations between strenuous outdoor job tasks and cross-shift or cross-harvest changes in kidney function observed in Central America support heat stress as an important contributor to disease progression in a population with pre-existing (and often subclinical) kidney injury not primarily due to heat.

NEPHROTOXIC WATER CONTAMINATION AS A CAUSE OF CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

Given the history of other endemic kidney diseases, exposure to nephrotoxic xenobiotics has unsurprisingly been proposed as a cause of disease. Metals, metalloids, and agrochemicals have all been described at high levels

in environmental sampling studies from the affected areas.^{45,50,51} Biomonitoring studies face substantial challenges including the short half-lives of agrochemicals, their seasonal and variable application, the difficulty in integrating lifespan, cumulative or time-average exposures, and reverse causation (where kidney injury of whatever cause is associated with high urinary concentrations of a number of metals). Nonetheless, high levels (either subjectively or when compared with international standards) of arsenic, nickel, silicon, and a number of pesticides have been described in populations at risk of or suffering from CKDu.^{52,53} A group of experts has also postulated that specifically sized silica nanoparticles released during the rice and sugarcane field burns (a common preharvest practice) may be nephrotoxic and have put forward supporting data from animal models.⁵⁴

One case-control study (nested in the Nicaraguan community-based cohort study described later) compared urinary concentrations of 12 metals and 12 pesticides among persons with rapidly changing kidney function and controls with stable kidney function, selected on the basis of 2 years of follow-up. In this study of modest size, but with one of the longest-reported follow-up periods, authors did detect pesticides or their metabolites in 25 cases and ~25 age- and sex-matched controls, but the concentrations were no different between cases and control groups (Fig. 83.8).⁵² In addition, case-control data and data within a case cohort are suggestive of a potentially higher bioburden of nickel.⁵³ Another study reported a greater likelihood of exposure to well water among persons with declining kidney function within a case cohort; well water in the CKDu-endemic region was graded as "hard" and contaminated with numerous pesticides.⁴⁵

INFECTION AS A CAUSE OF CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

The descriptions of episodes of acute febrile illness associated with kidney decline and histologic patterns of acute interstitial nephritis raise the possibility of a kidney-trophic infection such as leptospirosis, hantavirus, or as yet undescribed virus.⁵⁵ Although high rates of seropositivity to a number of pathogens have been described in both Central America and Sri Lanka, with some studies supporting modest differences in seroprevalence among case versus control populations,^{56,57} a large case-control study from Nicaragua did not find any substantive differences in leptospirosis or hantavirus seroprevalence among affected and unaffected individuals.⁵⁸

OTHER PROPOSED CAUSES OF CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

Phytotoxins and mycotoxins have been poorly studied in all regions, although absence of evidence of urologic malignancy (which is common in aristolochic acid nephropathy) suggest aristolochic acid is not responsible for the burden of disease observed. A higher proportion of affected than unaffected persons report family history of kidney disease in some epidemiologic studies,⁵⁹ and a single published whole-exome sequencing noted intriguing polymorphisms among persons with CKDu including in

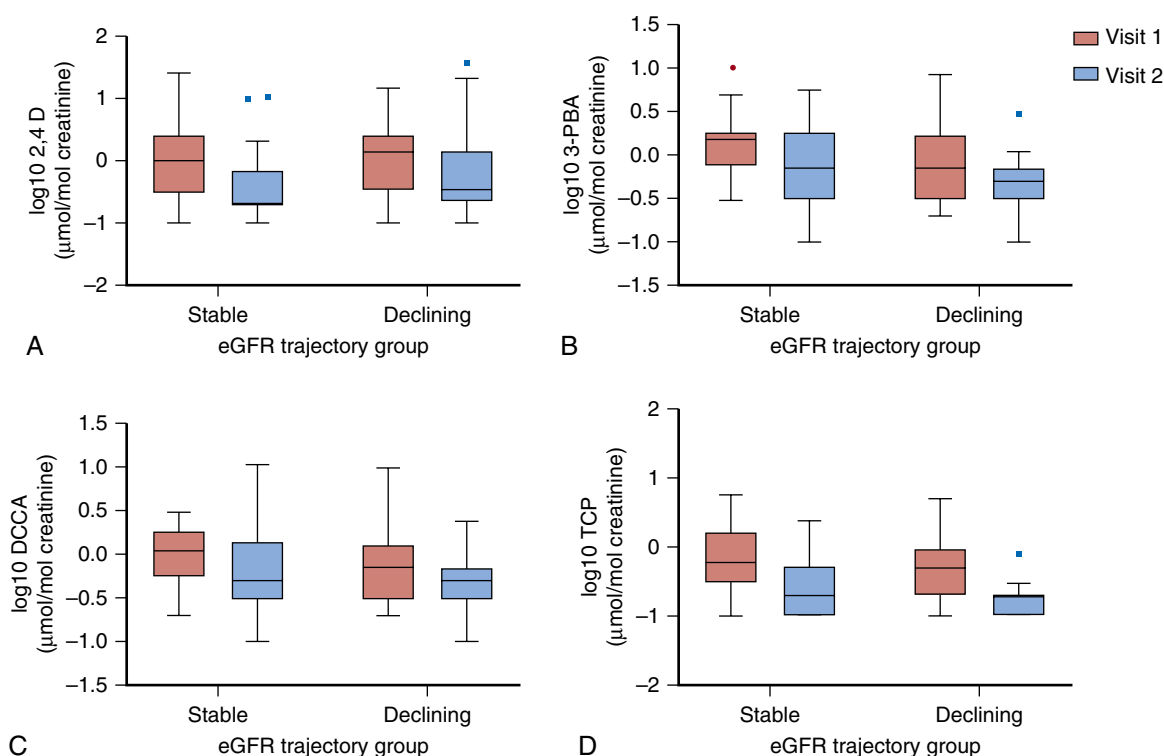


Fig. 83.8 Selected pesticide and pesticide metabolites urinary concentrations in a nested case-control study from Nicaragua. Creatinine-corrected pesticide concentrations at two baseline visits (6 months apart), comparing 25 cases with declining kidney function to those with stable kidney function over 2 years. Although both populations had detectable concentrations, no significant differences existed among persons with declining versus stable kidney function. (From Smpokou E-T, Gonzalez-Quiroz M, et al. Environmental exposures in young adults with declining kidney function in a population at risk of Mesoamerican nephropathy. *Occup Environ Med*. 2019;76:920–926.)

a gene encoding for a voltage-gated potassium channel localizing to the renal proximal tubule.⁶⁰

Some experts posit that CKDu is a multifactorial clinical syndrome, with more than one initiating and exacerbating factor. Although, of course, all diseases are multifactorial to a greater or lesser extent, it is worth noting that epidemic levels of disease typically occur when there is a change in a single factor (e.g., smoking and lung cancer; *H. pylori* and gastric ulceration). Furthermore, studies of different types will, by design, capture differing risk factors. For example, occupational studies will provide granular information on work exposures but will be less informative of home circumstances and typically not include women, who although less affected by CKDu than men, clearly can suffer from disease. The findings from selected studies conducted in the CKDu field are described as follows.

SELECTED STUDIES, COMPLETED OR ONGOING, INVESTIGATING CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

INTERNATIONAL COMPARISONS OF SURROGATES OF CKDU PREVALENCE

Despite the challenges in distinguishing CKDu from CKD, a large international effort has attempted to compare population-representative prevalence of CKDu using a pragmatic case definition, of (a single measure) eGFR <60

mL/min/1.73 m² in the absence of diabetes, hypertension (or treatment for hypertension), or heavy proteinuria in working-age adults. Termed the Disadvantaged Communities eGFR Epidemiology (DEGREE),³³ this collaboration aims at using pragmatic but robust approaches to produce comparable estimates prevalence of CKDu across regions. Currently published in preprint form, the preliminary reports from this collaboration, including 43 study centers across 14 countries are now available.⁶¹ These data demonstrate age-standardized prevalence of surrogate CKDu of 7% to 15% among males (aged 18–60 years old) in rural centers of the most impacted regions (i.e., in Central America, Sri Lanka, or southern India to date). Rates are typically lower in both women and urban populations. These figures compare with a prevalence of 0% and 0.5% of men in English and U.S. (combined urban and rural) surveys when the same case definitions are applied. Of note, however, a prevalence of <2% is observed using the same definitions among rural males in centers in Peru, Ecuador, Nepal, Thailand, and Malawi.

Care must be taken with interpreting larger studies, particularly national or subnational surveys where wide geographic sampling may obscure localized high rates of disease. Similarly, studies from sub-Saharan Africa and South Asia may underestimate prevalence as eGFR equations have been shown to overestimate the measured GFR in these regions^{62,63} (although estimates using cystatin C-based eGFR equations in the DEGREE publication did not change the overall observed pattern substantially). Nonetheless, the evidence to date suggests a high burden of CKDu is not

present in all regions with what might be thought to be typical risk factor profiles (e.g. hot and agricultural communities in low- or middle-income countries), in turn suggesting this constellation of risk factors may be necessary but not sufficient for disease evolution.

COMMUNITY-BASED FOLLOW-UP STUDIES

Several efforts are under way with medium- or long-term follow-up to attempt to ascertain incidence and risk factors for development and/or progression of kidney disease in the well-described hotspots.

One of the largest studies is based in Chinandega and Leon, Nicaragua, with the most recent report describing 771 participants followed for up to 7 years.⁶⁴ Participants were 18 to 30 years old at the time of recruitment, 478 were men (62%), and all had completed up to 10 study visits during follow-up. Among participants with eGFR >60 mL/min/1.73 m² at baseline, this study reported an incidence of CKD, defined by eGFR <60 mL/min/1.73 m² confirmed on two occasions, of 0.7%/year. CKD occurred exclusively among men, with an incidence rate 25-fold higher than age-comparable estimates from the United States. The research team also used statistical methods in an attempt to identify “early-stage disease” and found 1. a group who maintained normal eGFR (mean 115.1 [standard deviation 12.4] mL/min/1.73 m² at last visit) distinct from a group that started in the normal range but developed reduced eGFR, which was sustained to the end of follow-up (81.3 [standard deviation 24.0] mL/min/1.73 m² at last visit), and 2. although this latter group was composed of more men (15% of the men in the study), women were also represented (5% of the women in the study). Examining both “early-stage disease” and incident CKD using a “traditional” Cox proportional hazards approach, sugarcane work, and self-reported occupational sun exposure were associated with higher risk of the outcome. Measured and self-reported weight loss, symptoms of cramps, nausea, and diarrhea (as well as NSAID use) were also identified as risk factors for the onset of “early-stage disease.” The authors posited that although these latter symptoms could be related to heat, it was more likely that they reflected an inflammatory response due to an infectious or noninfectious toxin.

A second prospective study, called Kidney Protection Project, is ongoing in Sri Lanka’s Central Province, with 292 patients with reduced kidney function (eGFR 20–60 mL/min/1.73 m²) followed for 3 years thus far.²³ In this study, which followed 88% of all persons detected to have kidney dysfunction ($n = 330$) in a government screening campaign in the study region, the male predominance of CKDu was confirmed because 221 (76%) of the enrolled patients were men. A majority of the cohort had no detectable proteinuria on urine dipstick, and the rate of eGFR decline over 3 years was ~0.5 mL/min/1.73 m²/year. In exploratory analyses, the research team identified several geographic risk factors for >30% decline from baseline eGFR including drinking well water (at study baseline), living in the northern (higher elevation) study area, and living closer to an irrigation reservoir. Notably, the proportion of women experiencing declining kidney function was similar to men. Self-reported parameters of heat stress and agrochemical application were not associated with eGFR decline in this study. The research team posited that reservoirs in the region—which

hold pools of water that replenish only with rain—may concentrate groundwater contaminants. Given that case cohorts may be susceptible to reverse causation and survival bias, the research team has also instituted a parallel effort of a prospective study among “at-risk” persons with normal kidney function living in the same region. This At Risk for Kidney Disease in Sri Lanka study aims to recruit 600 participants, 300 men, and 300 women aged 18 to 45 years of age and in parallel obtain groundwater samples across two seasons. The study will perform time series integrated analyses from reservoir samples to characterize the water supply in a CKDu-endemic region.

The U.S. National Institutes of Health has sponsored a collaborative multisite project to investigate CKDu: Chronic Kidney Disease of Uncertain Etiology in Agricultural Communities (CURE) consortium.⁶⁵ Investigators from the United States, Central America, and India are collaborating on a prospective community-based cohort study with contemporaneous environmental sampling and biobanking, with participants drawn from two of the best-described hotspots. In addition to nephrologists and data scientists, this consortium has brought together investigators with expertise in environmental and occupational health, and laboratory measures of metals, metabolomics, genetics, and other advanced molecular phenotyping.

OCCUPATION-BASED STUDIES

Given the reported associations between CKDu and both residence in an agricultural community and agricultural work, there have been a range of studies, mainly conducted in Central America, which have focused on occupation either cross-sectionally or with short-term follow-up. Although study designs and locations vary, common themes can be drawn from this body of work. Firstly, studies demonstrate a higher burden of abnormal kidney function among participants in the most heat-exposed jobs or job roles.^{43,66,67} Secondly, those in the hottest job roles experience the greatest renal decline over the workday or work season.⁶⁸⁻⁷¹

One of the largest studies demonstrating this phenomenon was conducted in a sugarcane mill in Chinandega, Nicaragua, and recruited workers across four categories in descending order of physical strain: burned sugar cane cutting, green cane cutting, drip irrigation repair, and field supervisors or ancillary staff.⁷² Among the 545 participants recruited at the start of the study in October, at the beginning of the season, 427 had a repeat kidney function measurement performed 6 months later in April. Of note among the 98 persons without available measurements, 31 reported either sick leave or quitting their job due to elevated midharvest creatinine measures. Including these events in addition to persons with a change in serum creatinine >1.5 fold their baseline, the research team found that workers performing burned sugar cane cutting had ninefold higher odds of experiencing kidney injury during the harvest, relative to workers performing irrigation repair or other ancillary jobs. Similarly, trends were observed for eGFR change across harvest, and the authors concluded that physical occupational activity at hot temperatures was associated with kidney injury. Therefore interventions aimed at reducing the experience of heat stress during farming were imperative.

The Meso-American Nephropathy Occupational Study (MANOS) aims to expand occupational evaluation of kidney disease in El Salvador and Nicaragua.⁴⁴ In this study of 569 workers 18 to 45 years of age and working in sugarcane, brick, corn, plantain, and road construction, the prevalence of CKD (~14%) was again 8- to 10-fold higher than among age-comparable estimates in the United States. Sugarcane workers experienced the highest core temperature, likely driven by the metabolic demand of their work. Core temperature was, in turn, modestly higher among persons with eGFR <60 mL/min/1.73 m². Notably, however, in addition to sugarcane work, workers in corn and brick making had the second and third highest prevalence of CKD in this initial analysis. Further analyses inclusive of baseline biospecimens collected over 3 days of the work week and longitudinal follow-up for 2.5 years are under way.

INTERVENTIONS AND POLICIES

CENTRAL AMERICA

For years, the absence of adequate ESKD and/or CKD registries has obscured the real scope of the disease and thus the need for interventions. Nonetheless, a community-based local registry at the Bajo Lempa region in El Salvador shows that the incidence of kidney failure and mortality rates have remained high, since at least the beginning of the century and the beginning of record-keeping in this hotspot of MeN.⁷³

Both pesticides and heat stress have been linked to the disease by inhabitants of the communities and the scientific community.^{74,75} In El Salvador in 2010, a law regulating general risk prevention at the workplace was approved. This law includes guidelines for heat stress exposure and metabolic workload, but it is mainly enforced in the industry sector, whereas the MeN has been described among subsistence farmers and cooperatives. In 2013 the Legislative Assembly of El Salvador passed a law to ban 53 agrochemicals. The law was ultimately vetoed. Lastly, in Costa Rica in 2015, a law protecting workers against heat stress exposure directly related to CKD was also approved.

The U.S. Occupational Safety and Health Administration (OSHA) “Water. Rest. Shade” recommendations have also been implemented in select sugar cane companies across El Salvador and Nicaragua, aimed at reducing the incidence of AKI and CKD and the progression of CKD in these populations, with some promising preliminary results, mainly in the prevention of AKI.^{67,75,76} However the interventions are still limited to a select few sectors, without regular enforcement.

SRI LANKA

In the absence of a renal registry, the prevalence of CKD or CKDu is not certain in Sri Lanka. However, diseases of the urinary system were the fourth leading cause of hospitalization (1830 per 100,000 population) and the eighth leading cause of hospital deaths (14.4 per 100,000 population) in 2019. There are approximately 200 patients requiring dialysis per million population, overburdening the existing facilities, with nearly 90% receiving fewer than

two sessions of dialysis per week.⁷⁷ On this background, higher prevalence of CKDu in rural areas became a serious public concern. A Presidential task force to address CKDu was formed, and population screening programs were launched for early identification of affected individuals.⁷⁸ In parallel, renal services were improved while establishing sentinel sites for surveillance.

Several reports supported associations between CKDu and ground water contamination with agrochemicals.⁷⁹ Alarmed by these findings, several pesticides were banned, and usage of few pesticides was restricted to nonendemic areas. In 2021 in the mid of an ongoing economic crisis, a total ban on agrochemicals was implemented with the intention of changing the entire country to organic farming due to the perception that agrochemicals were harming population health. There were immediate counterprotests due to concerns about crop yield and food security. The decision was reversed after several months. Provision of safe water to endemic areas has been prioritized. As an interim measure, demineralized (reverse osmosis) water has been provided to CKDu-endemic areas. Rainwater harvesting and other water purification methods have been introduced with limited success. It has been reported that more than 80% of patients with a diagnosis of CKDu have changed water sources from well water to alternate sources.⁴⁵

INDIA

In response to the high prevalence of CKD in the Uddanam region, there have been several attempts to respond with policies, despite the acknowledgment that the specific cause of disease remains unknown. Media coverage has highlighted the disproportionate burden of CKD in the region, as well as the household and individual-level burden given limited access to dialysis care. Spurred by a community perception that the highest affected regions rely on contaminated groundwater, the state government is implementing a plan for bringing piped water to more households in the region.^{79,80} Furthermore, a new hybrid hospital-research facility is planned in collaboration with the Indian Council of Medical Research and the George Institute, India.⁸¹

INTERNATIONAL COLLABORATIVE EFFORTS

We have enumerated the challenges facing populations living in CKDu-endemic areas, researchers attempting to implement protocols for investigating cause(s), employers seeking effective interventions, and clinicians struggling to care for affected persons in underresourced areas. Yet all of these stakeholders are also collaborating and mobilizing in response to CKDu. Three examples include the International Society of Nephrology’s International Consortium of Collaborators on CKDu (i3C), the U.S. National Institutes of Health’s Chronic Kidney Disease of Uncertain Etiology in Agricultural Communities (CURE) consortium, and the Consortium for the Epidemic of Nephropathy in Central America and Mexico (CENCAM). The ISN’s i3C has collaborated with an international and multidisciplinary group to generate guidelines for research methodologies relevant to CKDu research and interventions.^{34,82,83} In addition, the ISN hosts an observatory



Fig. 83.9 A living observatory of ongoing and recently completed studies focused on chronic kidney disease of unknown etiology. The same colors indicate an international collaboration for the same study. While concentrated in Central American and South Asia, studies are ongoing in Africa and Middle East as well. (From ISN Observatory of CKDu. International Society of Nephrology, Belgium. Available at: <https://www.theisn.org/initiatives/what-is-ckdu/isn-observatory-of-ckdu>.)

of ongoing studies focused on investigating CKDu, which highlights not only the international collaborations being incubated as researchers attempt to identify the cause of the disease but also the multiple areas outside the established hot spots where others are evaluating whether a disease similar to CKDu exists (Fig. 83.9). The CURE consortium is implementing a cross-country protocol to evaluate environmental, occupational, and other risk factors for kidney function decline in agricultural communities.⁶⁵ CENCAM has ongoing engagement with representatives from the ministries of health across Central America and organizes biennial/triennial workshops bringing together nephrologists, occupational and environmental health specialists, epidemiologists, and public health professionals. Other collaboratives include industry-occupational health partnerships^{67,72,76} and public health agency initiatives.⁸⁴

CONCLUSIONS

Even as CKDu confounds scientists searching for potential etiologic factors, the hypothesized causes (e.g., occupational heat stress and agrochemical contamination) have motivated major sociopolitical policy changes. The consequences of these changes are difficult to discern due to variable data collection surrounding policy change and variable implementation efforts across regions. Nonetheless, the existence of CKDu has generated a range of international and multidisciplinary collaborations, interest in understanding population distribution of kidney function, and understudied risk factors for decline across diverse regions of the world, alongside attempts to integrate environmental epidemiology and occupational health with kidney disease research.

 Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references.

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